Managing Source Code Development Efficiently with Git

Version control systems (VCS) have made collaboration on large projects much simpler than before. This first article in a two-part series discusses Git, a very powerful and popular VCS. It looks at Git's important characteristics, its data structures and workflow, as well as some of its important basic commands and use cases.

With the advent of modern technologies, big, diverse teams working on large codebases became quite common. In hindsight, keeping track of code changes and different versions is very difficult. To solve this problem, version control systems were created. VCSs manage the changes made to the collection of information like documents, project code, etc. They track all the collaborative changes to a project and keep a record of all the previous versions.

The two major types of VCSs are: centralised and distributed.

The centralised system works on a client-server model with one centralised repository and multiple clients. Subversion is an example of a centralised VCS. The problem with this type of system is that you will always need to connect to the centralised repository to work on the project. So there's a single point of failure that can lead to availability issues, loss of data or the loss of code history. Distributed version control systems solve this problem. Distributed VCSs work on a peer-to-peer approach, whereby each developer has his/her own local repository. In this way, changes can be made to the code without needing any network connections. There is also no fear of data loss if one system fails, as it can be recovered from the other systems. Figures 1 and 2 show the schematic diagrams of a centralised and distributed VCS, respectively.

Git is an example of a distributed version control system. It was created by Linus Torvalds in 2005 for the development of the Linux kernel. Although described as 'the stupid content tracker' by its creator, Git soon became one of the most popular VCSs and is used equally by tech giants as well as the open source world.

What makes Git so popular?

At the time Git was being developed there were already a few version control systems in the market. It is Git's unique design and certain characteristics that resulted in it being so widely adopted compared to other VCSs. Some of those important characteristics are discussed here.